



Autonomous crypto incident response · system architecture

Google Cloud · Cloud Run · Firestore · Pub/Sub · Cloud Scheduler
Secret Manager · Vertex AI · Firebase Authentication
Agent Google ADK + Gemini 3.5 Flash on Vertex AI

1 · REQUEST PATH — CASE RETURNED IN ~50S

The victim submits a wallet. Nothing else is required: no transaction hash, no chain, no incident time.

Victim submits a wallet

Next.js web app on Cloud Run

CLOUD RUN

Firebase Authentication

ID token minted; every case is owner-scoped and cross-user access is denied

FIREBASE AUTH

FastAPI incident runtime

Verifies the token, then owns the investigation

CLOUD RUN

Incident discovery

Indexed wallet history returns every candidate outflow

ALCHEMY

Candidate ranking

Scores candidates and refuses to guess below 0.70 confidence, returning ranked finalists instead

Deterministic verification

The selected transaction is re-proved against the chain before anything is claimed

ETHEREUM + BASE JSON-RPC

Compromise assessment

Reads only verified evidence. Output is rejected if it attributes an entity the evidence cannot support

VERTEX AI · GEMINI 3.5 · ADK

2 · DEEP TRACE — OFF THE REQUEST PATH

Tracing a real drain runs for minutes, so it never blocks the response. The case is delivered as soon as its evidence stands.

DRAIN_REQUESTED published

One event per first-hop branch

PUB/SUB

Push subscription

Delivered back to the runtime over an authenticated internal endpoint

PUB/SUB → CLOUD RUN

Trace engine

Follows the funds hop by hop to depth 8: splits, residuals, swaps and bridges, with cross-chain continuation where the evidence resolves

Indexed continuation

A receipt cannot encode a contract-forwarded internal transfer. The index reveals it; RPC still proves the transaction

ALCHEMY + RPC VERIFY

Destination attribution

Vendored OFAC sanctions registry and curated bridge contracts, each carrying its own source

Firestore

Cases, branches, fund graph, timeline, processed events. Event claims are atomic, so redelivery cannot double-trace

FIRESTORE

3 · AUTONOMOUS MONITORING — NO HUMAN IN THE LOOP

Stolen funds sit still, then move. A dormant branch is not a finished one, so the agent keeps watching after the user has gone.

Cloud Scheduler

Fires every five minutes against an authenticated internal endpoint

*/5 * * * *

Dormant branch recheck

Every branch awaiting movement is re-queued for inspection

Movement detection

Indexed history for settled blocks, realtime stream for recent ones, and RPC verifies either before it counts

ALCHEMY · BITQUERY · RPC

Trace resumed automatically

Confirmed movement re-enters the trace engine through Pub/Sub. The graph, timeline and outcome update without anyone asking

PUB/SUB

Outcome and evidence package

Where the funds are, how much is unresolved, what is actionable, and what the victim can do next

Failure is reported as failure

A provider outage terminates a branch as evidence-retrieval-failed. It is never allowed to look like a finding about where funds are

EVIDENCE BOUNDARY — enforced in the data model, not in wording

Deterministic facts

Transactions, transfers, amounts, branch paths and attribution sources. Proved against the chain and stored before any model runs.

NEMESIS assessment

Gemini's reading of the compromise, carried separately with its own confidence and rejected if it names an entity the evidence cannot support.

Unknown and unresolved

Stated plainly. NEMESIS cannot freeze funds, recover funds, obtain account-holder identity, or compel any exchange or authority to act.